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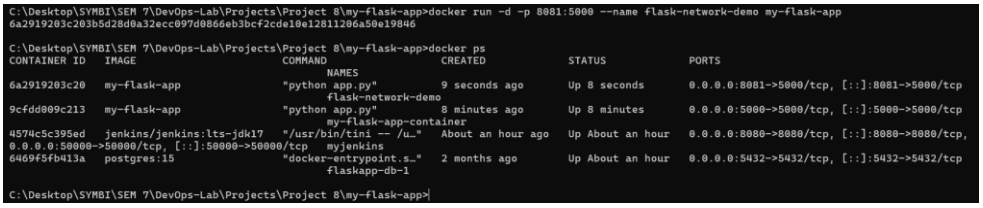
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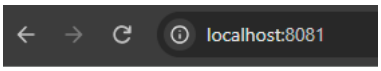
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 09

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yash Patel**
PRN No.	**23070122235**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Container Management & Networking
Theory:	Docker container management is the process of creating, running, monitoring, stopping, inspecting, and removing containers. Port mapping allows applications running inside a container to be accessed from the host machine by mapping a host port to a container port. Docker provides commands such as docker ps to view running containers, docker inspect to examine detailed container configuration and network settings, docker stop to gracefully stop a container, and docker rm to remove unused containers. These commands help efficiently manage the lifecycle of Docker containers and verify their network configuration and status. Commands Used • Created and started the Docker container in detached mode with port mapping: docker run -d -p 8081:5000 --name flask-network-demo my-flask-app • Listed all running Docker containers: docker ps • Stopped the running Docker container: docker stop flask-network-demo • Listed all Docker containers, including stopped containers: docker ps -a • Inspected the container's network configuration and settings: docker inspect flask-network-demo • Removed the stopped Docker container: docker rm flask-network-demo

	• Verified that the container had been removed: docker ps -a
Implementation:	**Step 1: Run a Container with Port Mapping** Started Docker Container — Started the Docker container named flask-network-demo in detached mode and mapped host port 8081 to container port 5000, allowing the application to run in the background. **Step 2: Access the Application via Your Browser** Verified Application Access — Accessed the Dockerized Flask application through http://localhost:8081 and verified that the application was successfully running inside the Docker container. **Step 3: List Running Containers** Listed Running Containers — Displayed all currently running Docker containers using the docker ps command and verified that the flask-network-demo container was running with port mapping from 8081 on the host to 5000 inside the container. **Step 4: Stop the Running Container** Stopped Docker Container — Stopped the running flask-network-demo container using the docker stop command, terminating the application gracefully. **Step 5: List Running and All Containers** Verified Container Status — Used the docker ps -a command to display all Docker containers and verified that the flask-network-demo container had stopped successfully and was in the Exited state. **Step 6: Inspect the Container's Network Settings** Inspected Container Network Settings — Used the docker inspect command to view detailed information about the flask-network-demo container, including its NetworkSettings and network configuration. **Step 7: Remove the Stopped Container** Removed Docker Container — Removed the flask-network-demo container using the docker rm command and verified its successful removal by confirming that it no longer appeared in the list of Docker containers. 



Hello, Docker!

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker stop flask-network-demo
flask-network-demo

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker ps -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED             STATUS                       PORTS
                                                                                                                         NAMES
6a2919203c20   my-flask-app                       "python app.py"          3 minutes ago       Exited (137) 30 seconds ago
                                                                                                                         flask-network-demo
9cfdd009c213   my-flask-app                       "python app.py"          12 minutes ago      Up 12 minutes                0.0.0.0:5000->5000/tcp,
[::]:5000->5000/tcp                                                                                                      my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u…"   About an hour ago   Up About an hour             0.0.0.0:8080->8080/tcp,
[::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp                                                     myjenkins
7319cd0ea244   grafana/grafana:latest             "/run.sh"                2 days ago          Exited (255) 2 hours ago     0.0.0.0:3000->3000/tcp
                                                                                                                         grafana
8d998a41987a   prom/prometheus:latest             "/bin/prometheus --c…"   2 days ago          Exited (255) 2 hours ago     0.0.0.0:9090->9090/tcp
                                                                                                                         prometheus
6422247b89a4e   cloud-negotiation-agent:v1         "python app.py"          2 days ago          Exited (255) 2 hours ago     0.0.0.0:5000->5000/tcp
                                                                                                                         cna-app
497d2dce6e7f   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   2 months ago        Exited (255) 8 weeks ago     127.0.0.1:32768->22/tcp,
 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   minikube
cb7d0620cccb   flaskapp-flaskapp                  "python app.py"          2 months ago        Exited (255) 8 weeks ago     0.0.0.0:5000->5000/tcp
                                                                                                                         flaskapp-flaskapp-1
6469f5fb413a   postgres:15                        "docker-entrypoint.s…"   2 months ago        Up 2 hours                   0.0.0.0:5432->5432/tcp,
[::]:5432->5432/tcp                                                                                                      flaskapp-db-1
aa5bb503eca7   flaskapp:latest                    "python app.py"          2 months ago        Exited (0) 2 months ago
                                                                                                                         eloquent_golick
e6dd76d6f617   flaskapp                           "python app.py"          2 months ago        Exited (0) 2 months ago
                                                                                                                         eager_mendel
21e44a730048   36926b4a7b0d                       "python app.py"          2 months ago        Exited (0) 2 months ago
                                                                                                                         stupefied_chandrasekhar

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker inspect --format="{{json .NetworkSettings}}" flask-netwo
rk-demo
{"SandboxID":"","SandboxKey":"","Ports":{},"Networks":{"bridge":{"IPAMConfig":null,"Links":null,"Aliases":null,"DriverOpts":null,
"GwPriority":0,"NetworkID":"06e72d78951ffa7a2c123adae37f6abec4025172dbd26d3504f5a95b8dbe4087","EndpointID":"","Gateway":"","IPAdd
ress":"","MacAddress":"","IPPrefixLen":0,"IPv6Gateway":"","GlobalIPv6Address":"","GlobalIPv6PrefixLen":0,"DNSNames":null}}}
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker rm flask-network-demo
flask-network-demo

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker ps -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED          STATUS                    PORTS
                                                                                                                     N
AMES
9cfdd009c213   my-flask-app                       "python app.py"          15 minutes ago   Up 15 minutes             0.0.0.0
:5000->5000/tcp, [::]:5000->5000/tcp                                                                                 m
y-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u…"   2 hours ago      Up 2 hours                0.0.0.0
:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp                                m
yjenkins
7319cd0ea244   grafana/grafana:latest             "/run.sh"                2 days ago       Exited (255) 2 hours ago  0.0.0.0
:3000->3000/tcp                                                                                                      g
rafana
8d998a41987a   prom/prometheus:latest             "/bin/prometheus --c…"   2 days ago       Exited (255) 2 hours ago  0.0.0.0
:9090->9090/tcp                                                                                                      p
rometheus
6422247b89a4e   cloud-negotiation-agent:v1         "python app.py"          2 days ago       Exited (255) 2 hours ago  0.0.0.0
:5000->5000/tcp                                                                                                      c
na-app
497d2dce6e7f   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   2 months ago     Exited (255) 8 weeks ago  127.0.0
.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   m
inikube
cb7d0620cccb   flaskapp-flaskapp                  "python app.py"          2 months ago     Exited (255) 8 weeks ago  0.0.0.0
:5000->5000/tcp                                                                                                      f
```

```
laskapp-flaskapp-1
6469f5fb413a   postgres:15                        "docker-entrypoint.s…"   2 months ago     Up 2 hours                0.0.0.0
:5432->5432/tcp, [::]:5432->5432/tcp                                                                                 f
laskapp-db-1
aa5bb503eca7   flaskapp:latest                    "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                     e
loquent_golick
e6dd76d6f617   flaskapp                           "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                     e
ager_mendel
21e44a730048   36926b4a7b0d                       "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                     s
tupefied_chandrasekhar

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```

Subject Faculty Name: Prof. Arijit Dutta